

OFFICE NETWORK BUILD VLANs

After setting up the basic network and understanding how Ethernet and switching work, the next challenge is separating the different departments within the office network. In a real office environment, allowing every department to share the same Layer 2 network can create unnecessary broadcast traffic and reduce network segmentation.

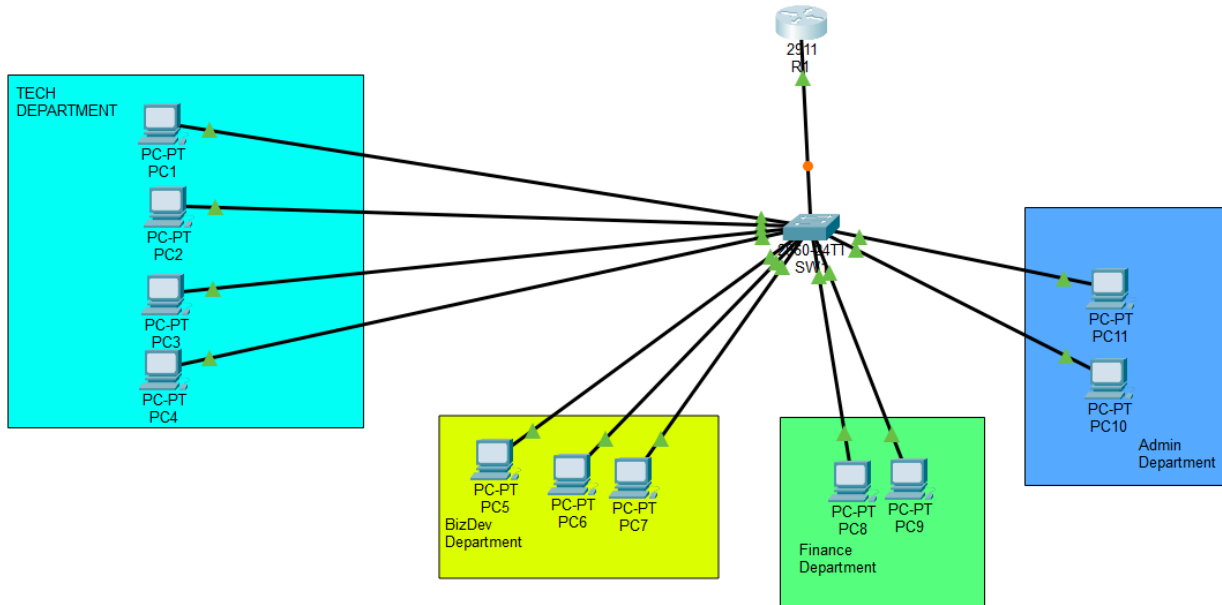
In this part of the project, I implemented Virtual Local Area Networks (VLANs) to logically separate the four departments of the simulated office network: Tech, Business Development, Finance, and Admin. This allows multiple departments to use the same physical switch while remaining in separate Layer 2 broadcast domains.

This section focuses on understanding VLAN creation, VLAN membership, access ports, and verification, establishing the foundation for later configurations such as trunking and inter-VLAN routing.

New IP Addresses have been set for the PCs to give way to the default gateway of each subnet

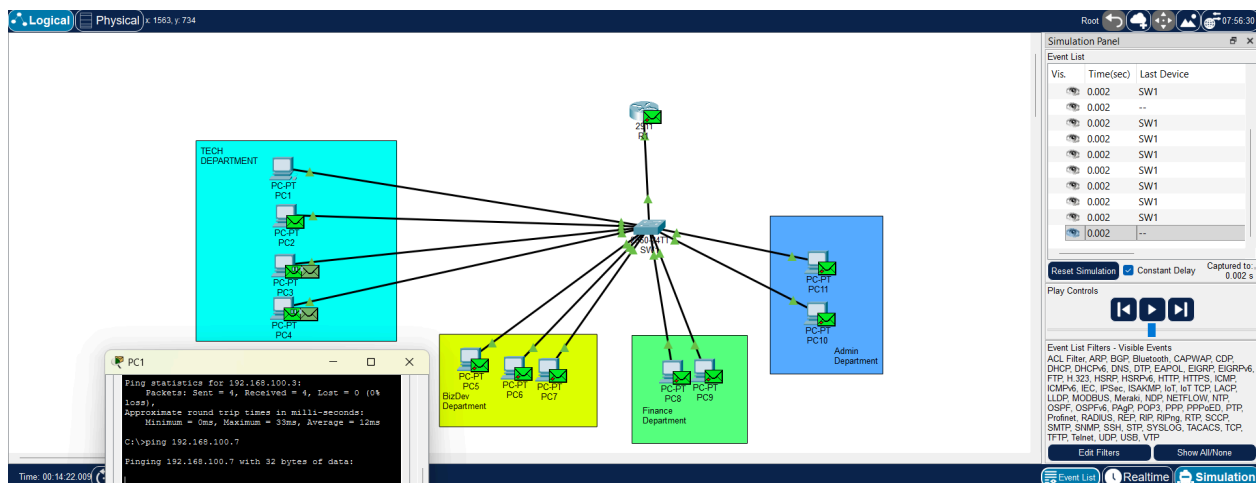
Office Network Build			
Device	IP	Subnet	Interface
R1	N/A	N/A	G0/0
PC1	192.168.100.2	255.255.255.248	Fa0
PC2	192.168.100.3	255.255.255.248	Fa0
PC3	192.168.100.4	255.255.255.248	Fa0
PC4	192.168.100.5	255.255.255.248	Fa0
PC5	192.168.100.10	255.255.255.248	Fa0
PC6	192.168.100.11	255.255.255.248	Fa0
PC7	192.168.100.12	255.255.255.248	Fa0
PC8	192.168.100.18	255.255.255.248	Fa0
PC9	192.168.100.19	255.255.255.248	Fa0
PC10	192.168.100.26	255.255.255.248	Fa0
PC11	192.168.100.27	255.255.255.248	Fa0

Current Topology



The Problem:

Without VLANs, all departments share the same Layer 2 broadcast domain. Broadcast traffic from one department can reach devices in other departments, increasing unnecessary network traffic and reducing network performance. It also provides less network security and segmentation, since devices from different departments remain within the same Layer 2 network. Below is an illustration of broadcast traffic without VLANs:



It can be seen that since traffic is still being sent to everyone even after subnetting. This is the problem VLANs can solve.

This will be the VLAN assignment of each department:

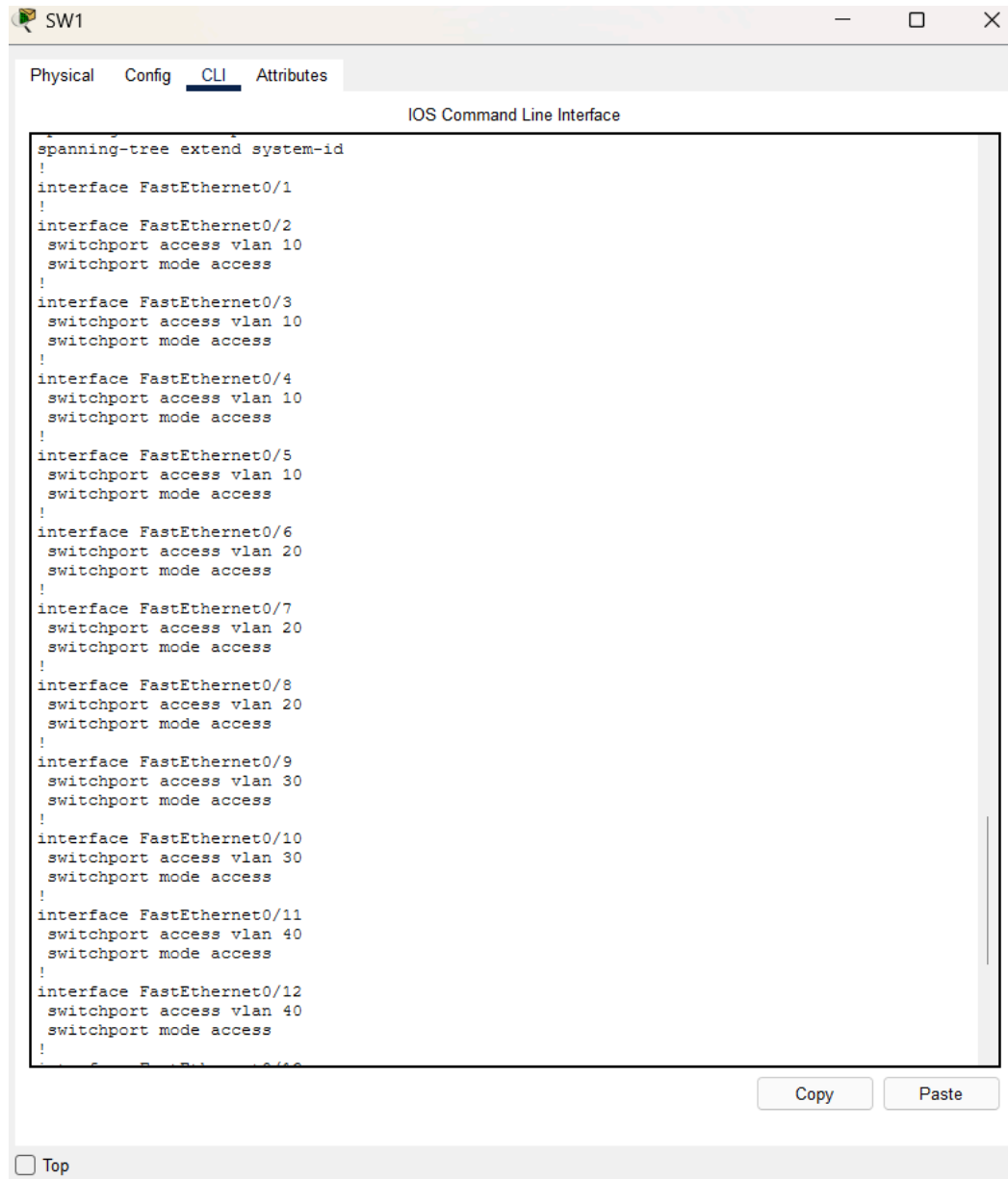
Tech Department - VLAN 10

Business Development DEpartment - VLAN 20

Finance Department - VLAN 30

Admin Department - VLAN 40

Startup Config (After VLAN Configuration):



The screenshot shows a window titled "SW1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The configuration text is as follows:

```
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/7
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/8
  switchport access vlan 20
  switchport mode access
!
interface FastEthernet0/9
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/10
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/12
  switchport access vlan 40
  switchport mode access
!
```

At the bottom of the window, there is a "Top" button and a "Copy" button. A "Paste" button is also visible at the bottom right of the configuration area.

show vlan brief output:

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW1#show vlan brief
```

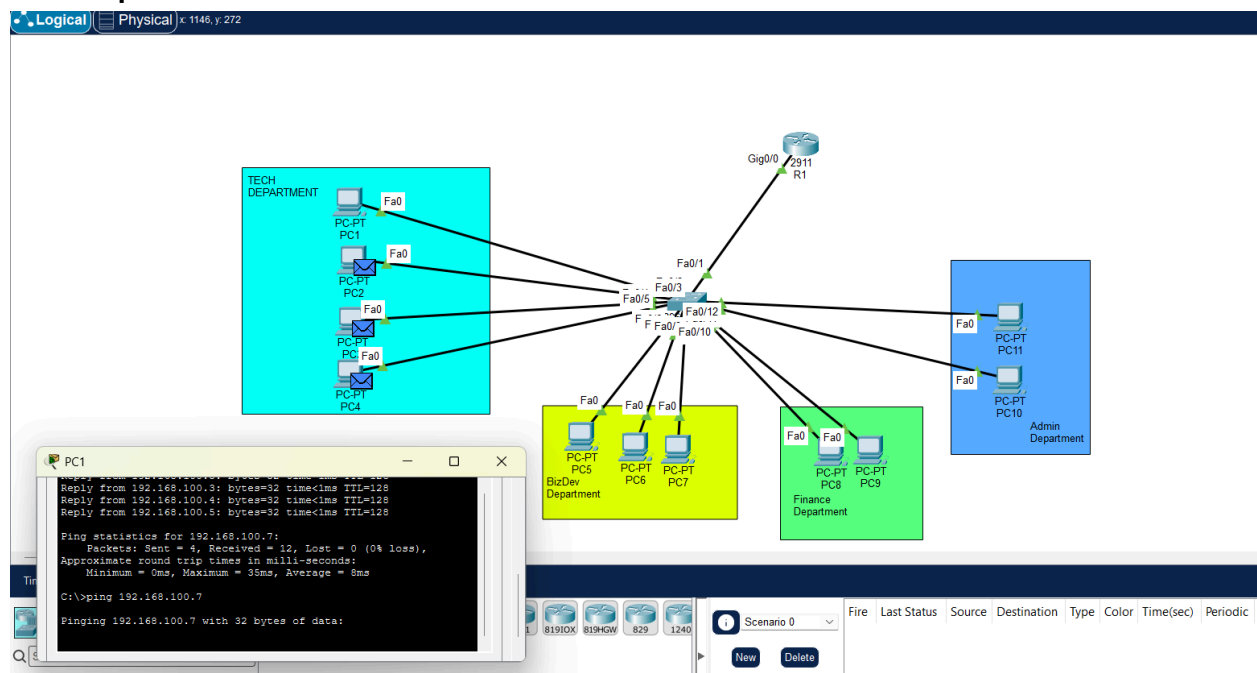
VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	TECH	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
20	BIZDEV	active	Fa0/6, Fa0/7, Fa0/8
30	FINANCE	active	Fa0/9, Fa0/10
40	ADMIN	active	Fa0/11, Fa0/12
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1#
```

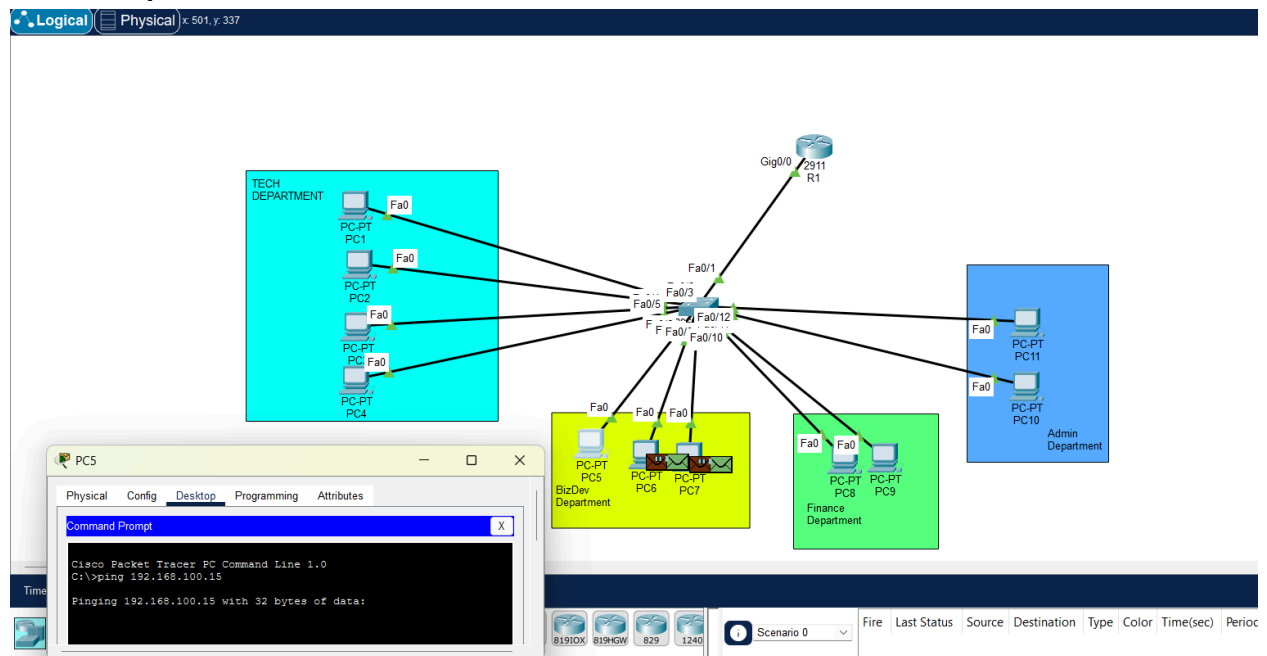
Copy

Paste

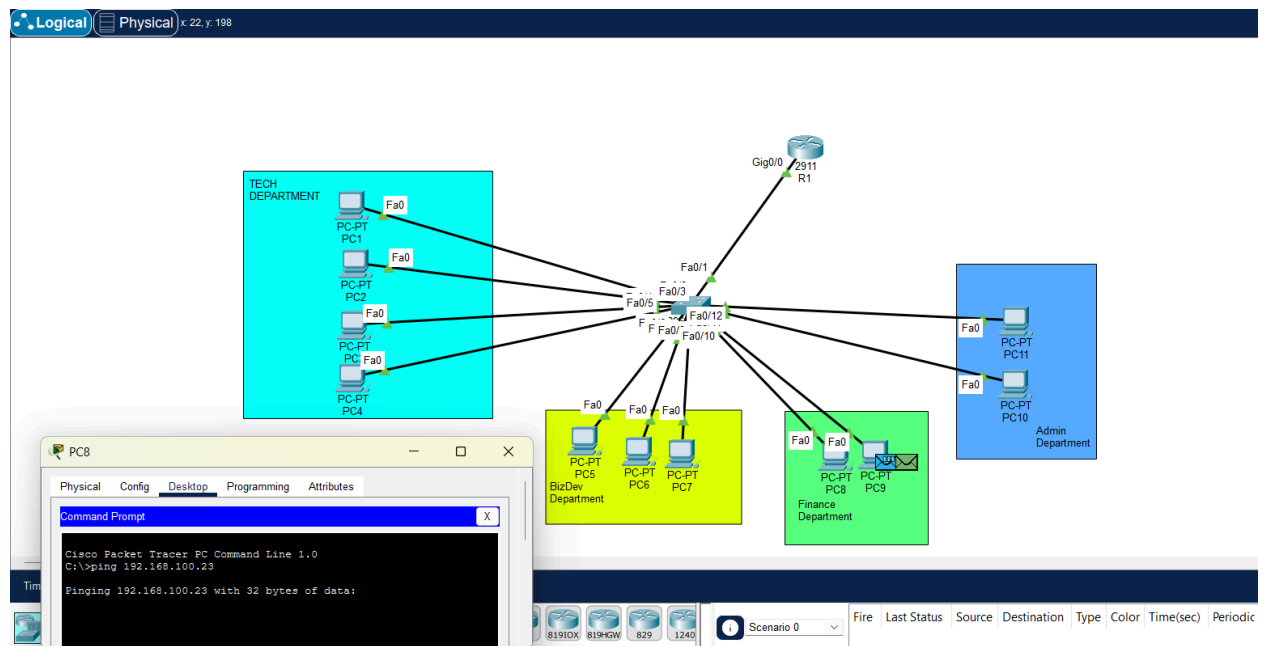
Tech Department Broadcast:



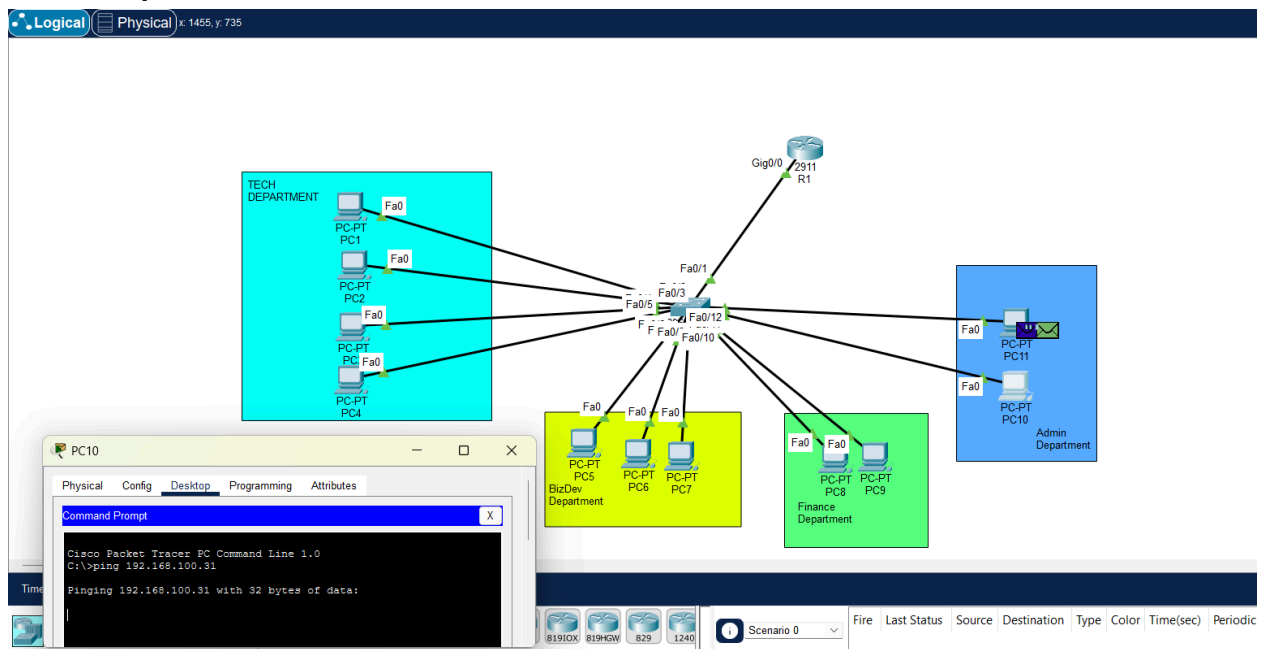
Bizdev Department Broadcast:



Finance Department Broadcast:



Admin Department Broadcast:



Conclusion:

The VLAN implementation successfully separated the office network into distinct Layer 2 broadcast domains for Tech, Business Development, Finance, and Admin. Devices within the same VLAN remained able to communicate, while broadcast traffic such as ARP was contained within its assigned VLAN and did not spread across the other departments. This demonstrated how VLANs can improve network segmentation, reduce unnecessary broadcast traffic, and provide a more organized foundation for network security and performance. The next stage of the project will build on this segmentation by using trunking to carry multiple VLANs across a single link.